

Natural Computing SoSe 2025

Exercise Sheet 4 (Solution) — June 24th, 2025

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1 Ising Model

Let

$$H'(\vec{\sigma}) = -7\sigma_1 + 5\sigma_2 - 15\sigma_3 - 3\sigma_4 + 12\sigma_1\sigma_4 + 6\sigma_1\sigma_2 + 8\sigma_2\sigma_3 + 2\sigma_1\sigma_3$$

be an energy function to be minimized, where $\vec{\sigma} \in \{-1, 1\}^4$. Recall that in the lecture we have covered the function to compute the energy $H(\vec{\sigma})$ of a spin vector $\vec{\sigma}$ given interaction matrix J and field vector $\vec{h}$ in an equivalent way to the following definition:

$$H(\vec{\sigma}) = \sum_{i=1}^N \sum_{j>i}^N \sigma_i \sigma_j J_{ij} + \sum_{i=1}^N h_i \sigma_i$$

(i) Give J and h to define the Ising model for $H' = H$.

$$J = \begin{pmatrix} 0 & 6 & 2 & 12 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \vec{h} = \begin{pmatrix} -7 \\ 5 \\ -15 \\ -3 \end{pmatrix}$$

(ii) Compute the energies $H(\vec{\alpha}), H(\vec{\beta})$ given J, h , and the following solution candidates

(a) $\vec{\alpha} = (1, 1, 1, 1)$

$$\begin{aligned} H(\vec{\alpha}) &= 1 \cdot 1 \cdot 6 + 1 \cdot 1 \cdot 2 + 1 \cdot 1 \cdot 12 + \\ &\quad 1 \cdot 1 \cdot 8 + 1 \cdot 1 \cdot 0 + \\ &\quad 1 \cdot 1 \cdot 0 + \\ &\quad (-7) \cdot 1 + 5 \cdot 1 + (-15) \cdot 1 + (-3) \cdot 1 \\ &= 8 \end{aligned}$$

(b) $\vec{\beta} = (-1, 1, -1, 1)$

$$\begin{aligned} H(\vec{\beta}) &= (-1) \cdot 1 \cdot 6 + (-1) \cdot (-1) \cdot 2 + (-1) \cdot 1 \cdot 12 + \\ &\quad 1 \cdot (-1) \cdot 8 + 1 \cdot 1 \cdot 0 + \\ &\quad (-1) \cdot 1 \cdot 0 + \\ &\quad (-7) \cdot (-1) + 5 \cdot 1 + (-15) \cdot (-1) + (-3) \cdot 1 \\ &= 0 \end{aligned}$$

(iii) Are the energies for either of the two solution candidates $\vec{\alpha}, \vec{\beta}$ already optimal? If not, find

$$\min_{\vec{x} \in \{-1, 1\}^4} H'(\vec{x}),$$

where H' is the energy function from task (i).

Neither $\vec{\alpha}$ nor $\vec{\beta}$ is optimal, as the corresponding energies to $\vec{x}_{\alpha} = (1, 1, 1, 1)$ and $\vec{x}_{\beta} = (-1, 1, -1, 1)$ are not optimal, i.e. minimal.

The minimal solution is $\vec{x}^* = (1, -1, 1, -1)$ with $H'(\vec{x}^*) = -48$.

$$\begin{aligned} H(\vec{x}^*) &= 1 \cdot (-1) \cdot 6 + 1 \cdot 1 \cdot 2 + 1 \cdot (-1) \cdot 12 + \\ &\quad (-1) \cdot 1 \cdot 8 + (-1) \cdot (-1) \cdot 0 + \\ &\quad 1 \cdot 1 \cdot 0 + \\ &\quad (-7) \cdot 1 + 5 \cdot (-1) + (-15) \cdot 1 + (-3) \cdot (-1) \\ &= -48 \end{aligned}$$

2 Ising to QUBO

Let again

$$H'(\vec{\sigma}) = -7\sigma_1 + 5\sigma_2 - 15\sigma_3 - 3\sigma_4 + 12\sigma_1\sigma_4 + 6\sigma_1\sigma_2 + 8\sigma_2\sigma_3 + 2\sigma_1\sigma_3$$

be an energy function to be minimized, where $\vec{\sigma} \in \{-1, +1\}^4$ is our solution candidate. To compute the energy $H(\vec{\sigma})$ of a solution vector $\vec{\sigma}$, given interaction matrix J and field

vector $\vec{h}$, we recall the following definition:

$$H(\vec{\sigma}) = \sum_{i=1}^N \sum_{j>i}^N \sigma_i \sigma_j J_{ij} + \sum_{i=1}^N h_i \sigma_i$$

Convert the optimal Ising solution $[1, -1, 1, -1]$ to a QUBO solution and explain why we need the conversion $\sigma_i = 2x_i - 1$ for all i , if we want to keep the energy values consistent.

The same type of formula, $\sum_{i=1}^N \sum_{j>i}^i x_i x_j J_{ij} + \sum_{i=1}^N h_i x_i$, works for Ising and QUBO, but for Ising $x_i \in \{-1, 1\}$ and for QUBO $x_i \in \{0, 1\}$ for any i .

Without the conversion, the returned energy values would not produce the same landscape (and thus the same optimization problem instance). For example given $\vec{\beta} = (1, -1, 1, -1) \implies \vec{x} = (1, 0, 1, 0)$, we could naively and wrongly combine J and $\vec{h}$ into a matrix O and compute...

$$\begin{aligned} H_{\text{“QUBO”}}(\vec{\beta}) &= x^T O x = \begin{pmatrix} 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -7 & 6 & 2 & 12 \\ & 5 & 8 & 0 \\ & & -15 & 0 \\ & & & -3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\ &= -7(1) + 5(0) - 15(1) - 3(0) + 6(1)(0) + 2(1)(1) + 12(1)(0) + 8(0)(1) \\ &= -20 \neq -48 = H(\vec{\beta}) \text{ (see Ising Exercise)} \end{aligned}$$

but get a different resulting energy. Instead, we need to convert J and $\vec{h}$ into the isomorphic QUBO matrix Q by substituting every σ_i with $2x_i - 1$ according to the conversion rule. We are thus computing...

$$\begin{aligned} H(\vec{\sigma}) &\stackrel{!}{=} H_{QUBO}(\vec{x}) = -7(2x_1 - 1) + 5(2x_2 - 1) - 15(2x_3 - 1) - 3(2x_4 - 1) \\ &\quad + 12(2x_1 - 1)(2x_4 - 1) + 6(2x_1 - 1)(2x_2 - 1) \\ &\quad + 8(2x_2 - 1)(2x_3 - 1) + 2(2x_1 - 1)(2x_3 - 1) \\ &= -14x_1 + 7 + 10x_2 - 5 - 30x_3 + 15 - 6x_4 + 3 \\ &\quad + 48x_1x_4 - 24x_1 - 24x_4 + 12 + 24x_1x_2 - 12x_1 - 12x_2 + 6 \\ &\quad + 32x_2x_3 - 16x_2 - 16x_3 + 8 + 8x_1x_3 - 4x_1 - 4x_3 + 2 \\ &= -54x_1 - 18x_2 - 50x_3 - 30x_4 \\ &\quad + 24x_1x_2 + 8x_1x_3 + 48x_1x_4 + 32x_2x_3 + 48 \end{aligned}$$

... from which we can read ...

$$Q = \begin{pmatrix} -54 & 24 & 8 & 48 \\ & -18 & 32 & 0 \\ & & -50 & 0 \\ & & & -30 \end{pmatrix}$$

With this correct Q matrix we can compute the energy $H_{QUBO}(\vec{x}) = x^T Q x = -96$. Note that this conversion yields a constant *offset* (in this case, +48) by which the respective energies for Ising and QUBO differ. However, since the constant offset only shifts the energy landscape but does not affect its shape, it is often left out in practice.